Poor Sleep Habits Among Adults: Using Technology to Promote Healthy Sleep

A report on how to combat poor sleeping habits

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Introduction

Sleep plays a very important role in our lives. It promotes good health and replenishes your energy for the next day you are awake. In this stage, your body is working hard enough to support all its functions, especially your brain. A good night's rest can provide many benefits, such as repairing and maintaining the brain's ability to work properly while clearing out the waste and toxins that accumulate during the day. It also strengthens your memory, helps you more easily process new information, and enhances your capabilities and overall mental health.

Adults are generally recommended to get 7–9 hours of sleep per night. However, insufficient or poor-quality sleep remains a common problem. In the United States, 30.5% of adults reported sleeping less than 7 hours on average in 2024. The same report found that 15.4% of adults had trouble falling asleep and 18.1% had trouble staying asleep. (CDC, 2024) Poor sleep habits include not only getting too few hours of sleep but also behaviors and routines that interfere with adequate, consistent sleep. Adults may develop unhealthy sleep patterns because of demanding work or school schedules, stress, late-night entertainment, excessive smartphone use, irregular bedtimes, caffeine consumption, long daytime naps, or other lifestyle factors. According to the National Heart, Lung, and Blood Institute (NHLBI), several behavioral factors may trigger sleep disturbances. These include shifting one's rest schedule, taking extended naps during the day, consuming caffeine, experiencing high levels of stress, and interacting with digital screens shortly before retiring for the night.

The Problem

Especially among adults, this is a relevant problem in how society treats its employees and their time, because adults often have schedules that conflict with their need for adequate rest. Work responsibilities, academic requirements, household duties, social activities, and digital entertainment can cause people to delay bedtime. Some individuals may also believe that sacrificing sleep makes them more productive, but insufficient sleep can instead contribute to tiredness and reduced daytime functioning. Poor sleep habits can have consequences beyond just feeling tired in the morning. A poor sleep schedule has been linked to several health problems, such as high blood pressure, obesity, low immune function, which can make a person vulnerable to such conditions such as diabetes, heart disease, and respiratory infections, such as the flu. (Watson, 2025)

Many adults have difficulty establishing and maintaining a regular sleep routine. They may sleep at different times each night, stay awake for long periods using their phones or computers, consume caffeinated beverages late in the day, or fail to give themselves enough time for sleep. These behaviors can disrupt the body's normal sleep-wake cycle, making it more difficult to fall asleep or maintain sleep. Another important aspect of the problem is the lack of awareness regarding personal sleep patterns. A person may know that they are tired but may not recognize that their irregular bedtime, frequent late-night screen use, or caffeine consumption is contributing to the problem. Without a simple method for recording and reviewing these behaviors, it can be difficult for individuals to identify patterns and determine which habits to change.

Possible Solutions

One of the possible solutions is using a sleep tracking app to recognize the poor sleeping habits.With using a a mobile-based application designed to help adults recognize poor sleep habits and develop healthier sleep routines. The system combines sleep tracking, personalized reminders, sleep-hygiene education, and progress monitoring to provide users with practical support for improving their sleep quality. Rather than simply recording how long a person sleeps, SleepSmart focuses on the behaviors and environmental factors that may influence sleep. The purpose of these apps is to make healthy sleep practices easier to understand and incorporate into everyday life. Sleep hygiene refers to habits and environmental conditions that support consistent and good-quality sleep. Improving sleep hygiene can involve maintaining a regular sleep schedule, creating a comfortable sleeping environment, establishing a relaxing bedtime routine, limiting electronics and stimulants before bed, and getting adequate exposure to natural light during the day.

Other solutions such as avoid electronic devices before bedtime. Smartphones, computers, tablets, and televisions can encourage people to remain awake longer and expose them to bright light.

Another important factor is creating a comfortable sleeping environment. A bedroom that is dark, quiet, and comfortable can make it easier to sleep. Individuals can reduce unwanted light by using curtains or blinds, minimize disruptive sounds, and maintain a comfortable room temperature.

Maintaining a consistent sleep schedule is also important. Going to bed and waking up at approximately the same time every day can help establish a regular sleep-wake rhythm. Large differences in sleep times across days may make it more difficult for the body to maintain a predictable routine.

Caffeine and alcohol consumption should also be considered when improving sleep habits. Caffeine is a stimulant that can remain in the body for several hours and may interfere with the ability to fall asleep, particularly when consumed later in the day. The NHLBI recommends avoiding caffeine close to bedtime.

Conclusion

Poor sleep habits can negatively affect adults’ health, mood, concentration, productivity, and overall well-being. Practicing good sleep hygiene, such as maintaining a regular sleep schedule, limiting screen time and caffeine before bed, creating a comfortable sleep environment, and getting enough sleep, can help improve sleep quality. However, maintaining these habits consistently can be difficult. The proposed SleepSmart: A Smart Sleep Management System can support adults through sleep tracking, reminders, personalized goals, and progress reports. By combining healthy sleep practices with technology, the system can encourage users to develop more consistent and healthier sleep routines.

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